AI AGENT CAPSTONE · WEEK 4

Project Documentation Assistant

RAG, one safe tool and a LangGraph workflow over my own internship documentation

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Quick answers from my own documentation, with exact arithmetic

It should

- Answer from 7 documentation files
- Compute differences and ratios of numbers found in them
- Decline when information is missing or out of scope
- Reject invalid input

It should not

- Answer general-knowledge questions
- Invent numbers
- Browse the web
- Edit files or run code

One graph: 8 nodes, 4 branch points

Node	Job
validate_input	Text with at least 3 characters
retrieve	FAISS search, k = 4, distance up to 1.2
plan_calculation	Gemini proposes one expression, or NONE
calculate	Number check, then a safe calculator
generate	Gemini answers from the chunks and the result

Decline paths: invalid_reply · no_answer · cannot_verify

Every run records the path of nodes it took.

One question, end to end

By how many points did the Keras spam recall exceed the baseline recall?

Step	What happened
retrieve	4 chunks; rank 1 from root_readme.md (score 0.588)
plan_calculation	Expression: 0.86 - 0.71
calculate	Both numbers found in the chunks; result 0.15
generate	"...exceeded the baseline recall by 0.15 points"

Known issue: the gap is 15 percentage points; the answer says "points".

Built to decline instead of guess

Safe calculator

Parsed with ast, no eval. Only numbers and + - * / are allowed.

Number check

Every number in a calculation must appear in the retrieved text (100 is allowed).

3 decline paths

Invalid input, no relevant chunk, calculation not verified.

Tests: calculator 17, number check 12, branches 6 (stand-in model). All passed.

10 tests, expectations written before running

- Pass: T1, T2, T7, T8
- Pass with notes: T3, T4
- Safe but unhelpful declines: T5, T6
- Tool not reached: T9
- T10: failed, fixed, passed

0.1 s

when the graph stops before Gemini

2.6-11.6 s

when Gemini runs (1 to 2 runs per question)

Failures found, and what is still open

Finding	Status
T10: the constant 100 blocked a valid percentage	Fixed, re-tested
Unit wording: "0.15 points"	Open
Offer Letter ID chunk dropped at distance 1.414	Open
Approximate values stated as exact	Open
Ambiguous question gets a bare decline	Open
Baseline not named (4,136 vs 3,515 train)	Open

Limits and next steps

Limits

- Distance limit 1.2 is untuned
- 7 files; 1 to 2 runs per question
- Fresh clone tested on one Windows machine

Next

- Tune the limit with more questions
- Ask a clarifying question
- Skip planning when no numbers

Repository: github.com/Rakshithamrakshi/skillset-go-aiml-internship-rakshitha-m (Week-4/Task-4.4-AI-Agent-Capstone) · Live demo: [link to add]